Bioreactor Output : Algae Biomass Production and CO_2 Capture

Tuesday, June 17, 2025 10:04 AM

Expected Daily Dry Chlorella Vulgaris Biomass Production

Goal: To find out how many kilograms of dry Chlorella Vulgaris biomass the bioreactor needs to produce daily to successfully remove targeted amounts of Nitrogen and Phosphorus from the wastewater.

→ Daily wastewater flow (Q): 2,400 L/day
→ Assumed Influent Phosphorus Concentration (P_{in}) : 7 mg/L
→ Assumed Influent Nitrogen Concentration (N_{in}) : 40 mg/L

The standard algae composition values (for Chlorella Vulgaris):
- Nitrogen (N) content in Dry Algae : 8%
- Phosphorus (P) content in Dry Algae : 1%

→ Step 1: Calculating the daily mass of Phosphorus (P) to be removed
This calculates the total mass of phosphorus, in milligrams, that needs to be extracted from the wastewater daily based on my targeted removal efficiency.

$$P_{removed}\ (mg\ P/day) = P_{in}\ (mg/L) \times Q\ (L/day) \times \left[\frac{Target\ Removal\ Efficiency}{100}\right]$$

$$= (7\ mg/L) \times (2400\ L/day) \times \frac{95}{100}$$

$$= 15{,}960\ mg\ P/day$$

→ Step 2: Calculate the daily Mass of Nitrogen (N) to be removed
Similar to step 1 this step calculates the total mass of nitrogen, in milligrams, that needs to be removed daily.

$$N_{removed} = 40 mg/L \times 2400\ L/day \times \frac{95}{100}$$

Nitrogen removed = 91,200 mg N/day

→ Step 3: Calculating Daily Dry Biomass Required based on Phosphorus Removal

Biomass_P (mg dry algae/day) = P removed (mg P/day) / (P content in dry algae)

Biomass_P = 15,960 mg P/day / 0.01 (since P content is 1%)

Biomass_P = 1,596,000 mg dry algae/day

Divide 1.596 by 1,000,000 (1 kg = 1,000,000)
$= \frac{1{,}596{,}000}{1000\ 000} = 1.596$ kg algae/day

→ Step 4 : Calculate Daily Dry Biomass Required Based on Nitrogen (N) Removal

Biomass_N (mg dry algae/day) = N removed / (N content in dry algae)

Biomass_N = 91,200 mg N/day / 0.08
= 1,140,000 mg dry algae/day

Converted to kg:
1,140,000 mg / 1,000,000
Biomass_N = 1.140 kg dry algae/day

Step 5: Determining Final Expected Daily Dry Chlorella vulgaris Biomass Production

Biomass_P = 1.596 kg/day
Biomass_N = 1.140 kg/day
> choosing the bigger value!

Final Result for daily dry Chlorella vulgaris Biomass Production:
⇒ 1.596 kg dry algae/day

What this means:
Based on the calculated nutrient removal requirements (from an influent of 7mg/L P and 40mg/L N with 95% removal efficiency) within the 15,600 Liter bioreactor, the system is expected to produce approximately 1.6 kg of dry Chlorella Vulgaris algae biomass per day.

Quantifying CO_2 Capture

Objective : To calculate how much CO_2 the bioreactor is estimated to capture daily, based on the algae biomass produced.
Assumption : Carbon content in dry algae : Approx. 50% (standard average value)
Step 1 : Calculate the mass of carbon contained in the algae biomass
This step the amount of pure carbon locked up in dry algae biomass produced each day.

Mass of Carbon (kg C/day) = Expected daily dry algae biomass (kg dry algae/day) X Carbon content in dry algae in decimal

= 1.596 kg dry algae/day x 0.50 mass of carbon = 0.798 kg C/day

Step 2: Convert Mass of Carbon to Mass of CO_2 captured
This step uses the molecular weights of carbon (C) and carbon dioxide (CO_2).

Molecular Weight:
Carbon (C) ≈ 12 g/mol
Carbon dioxide ≈ 44 g/mol
(12 + 2(16)) = 44g/mol

This means : every 12 units of carbon → come from 44 units of CO_2

Formula: Mass of CO_2 captured (kg CO_2/day) = Mass of Carbon (kg C/day) x (44/12)

$= 0.798 \times (44/12) = 2.9\ 6$ kg CO_2/day

* The key assumptions mentioned in the calculations are stated in week 2 reference file (Assumptions_day 5-6) for more information.